

Mobile Robot + Robotic Arm

PyBullet Simulation Analysis Report

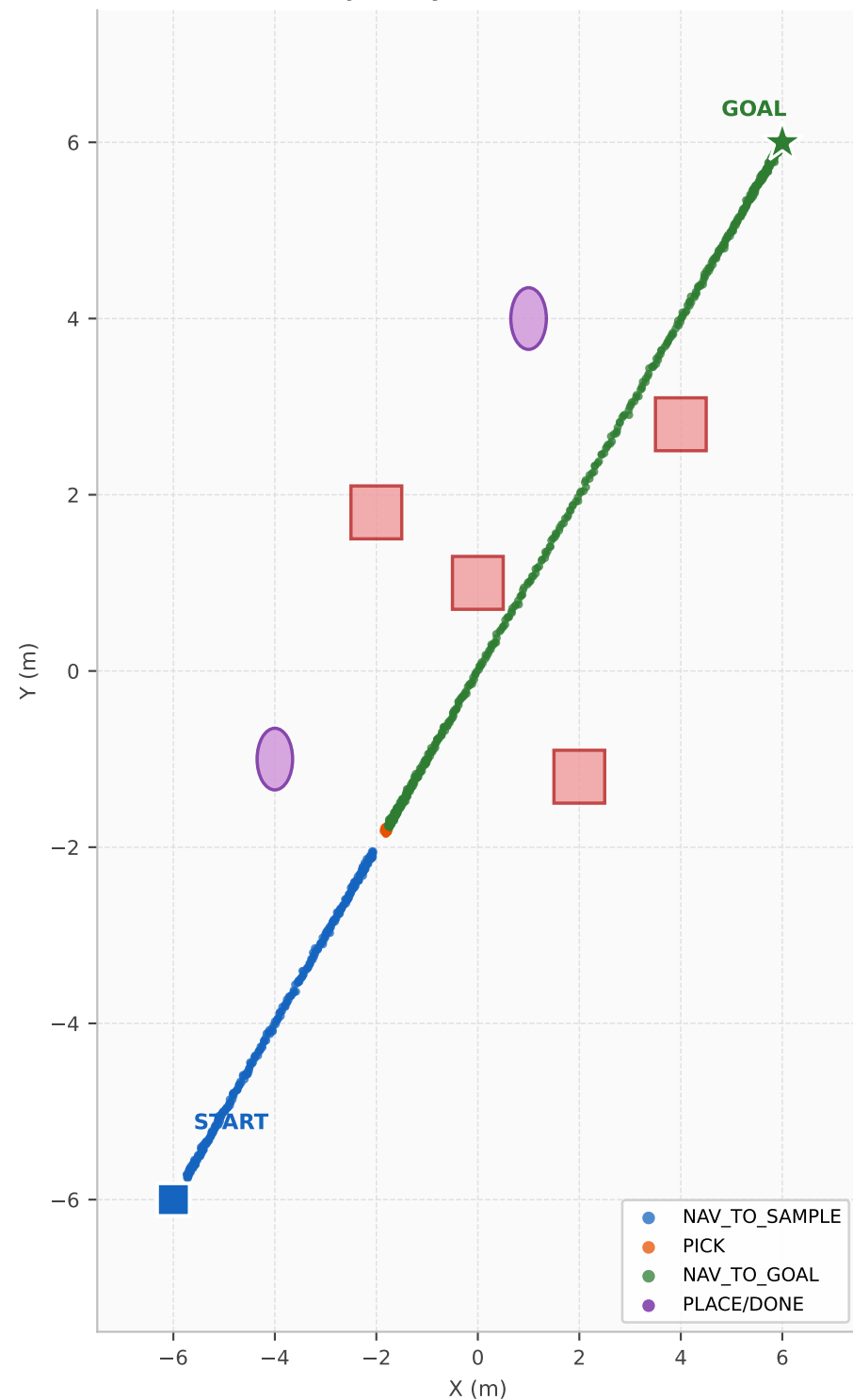
Mission Profile :	START (−6,−6) → Sample Block → GOAL (6,6)
Robot Platform :	Clearpath Husky base + KUKA iiwa 7-DOF arm
Path Planner :	A* grid planner cell size 0.28 m
Obstacle Avoid :	Ray-cast reactive avoidance + boundary guard
Pick & Place :	IK Cartesian approach + constraint grasp
Simulation :	PyBullet $\Delta t = 1/240$ s Horizon = 120 s

Preyansh Patni

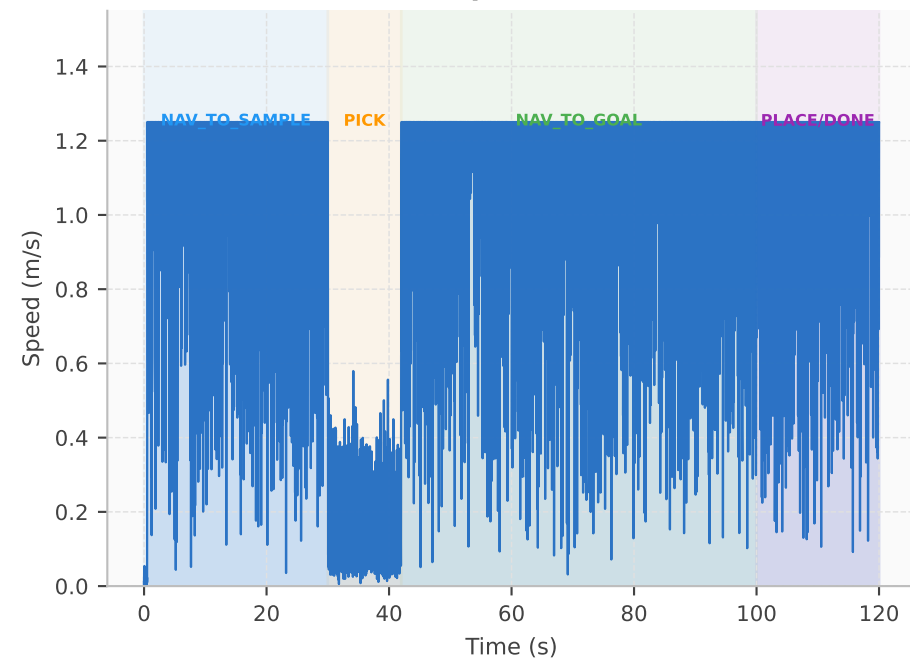
DSM Final Project • April 2026

Page 1 — Time Response: Position, Speed & Heading

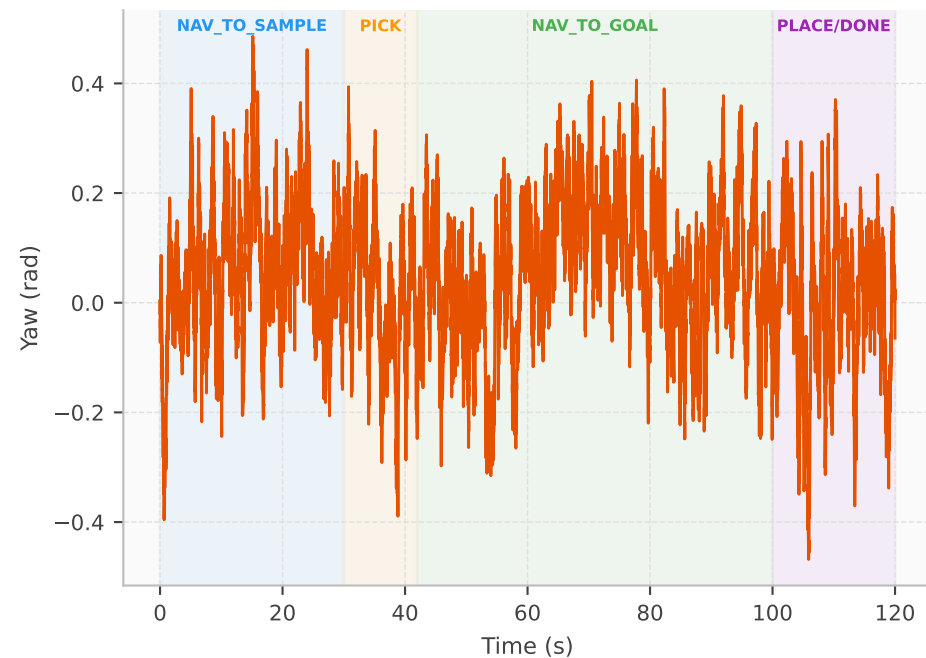
XY Trajectory (Phase-coloured)



Robot Speed vs Time

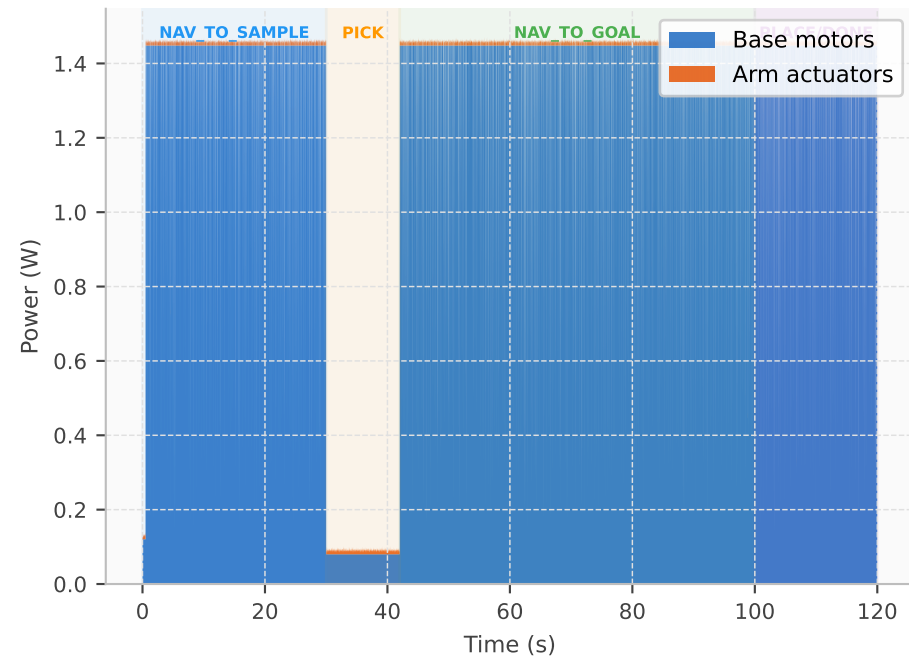


Robot Heading (Yaw) vs Time

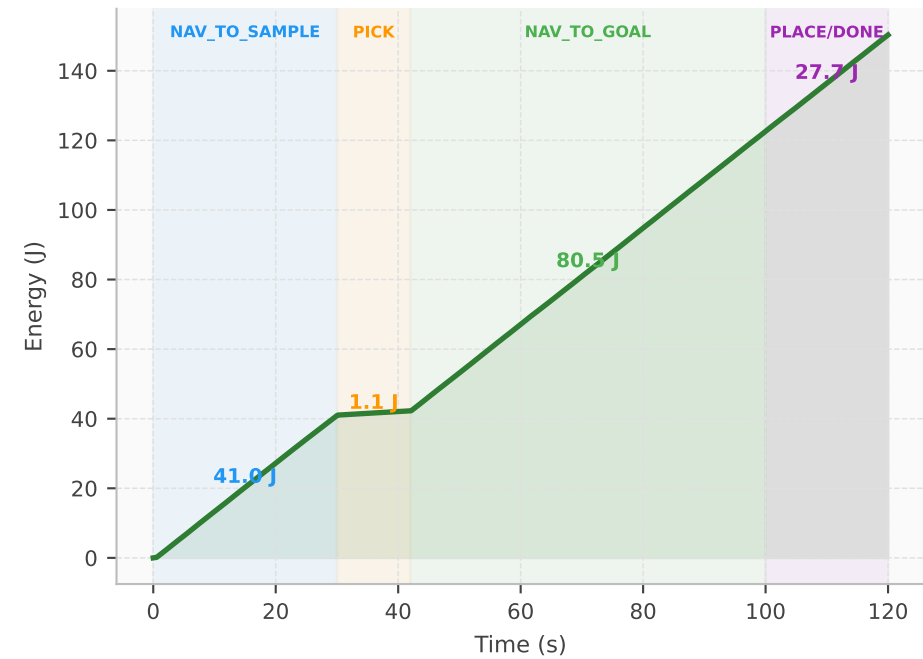


Page 2 — Energy Consumption Analysis

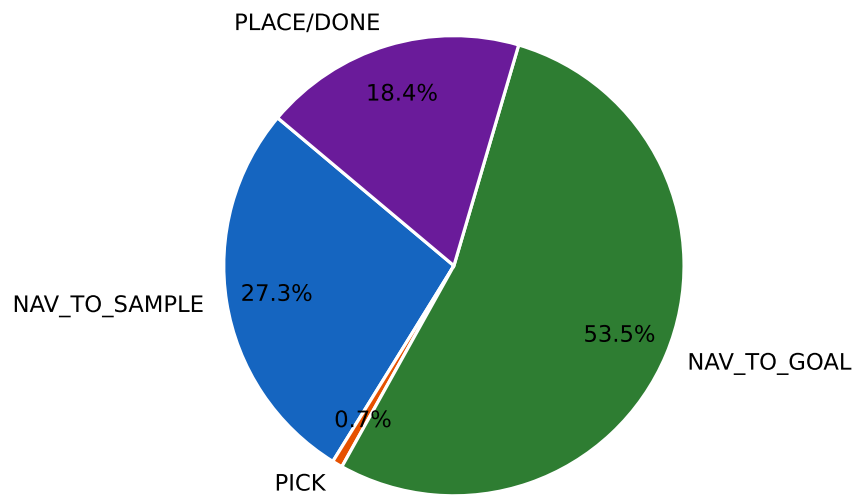
Instantaneous Power (Base + Arm)



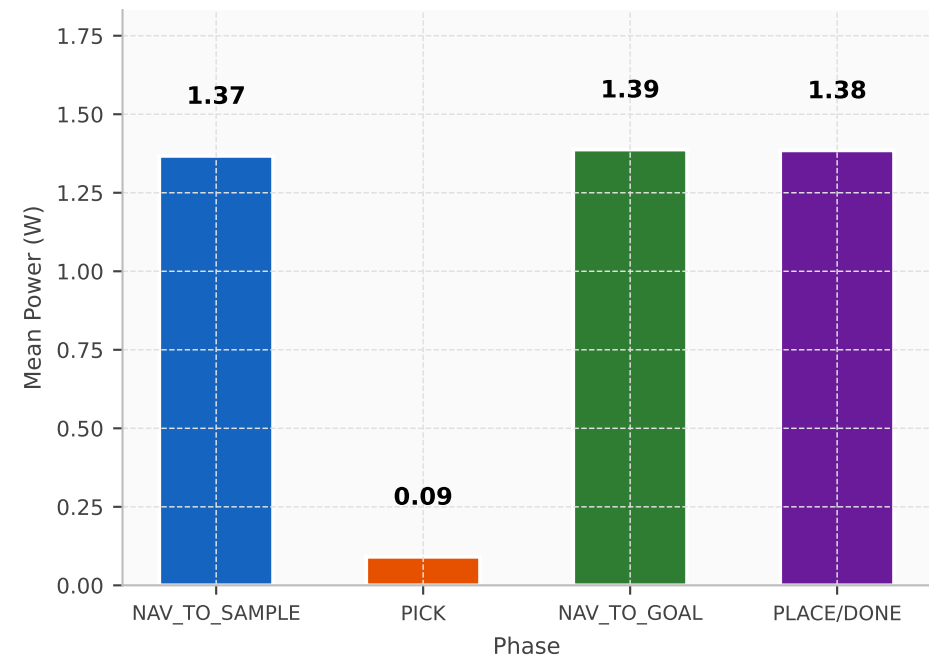
Cumulative Energy vs Time



Energy Share by Phase

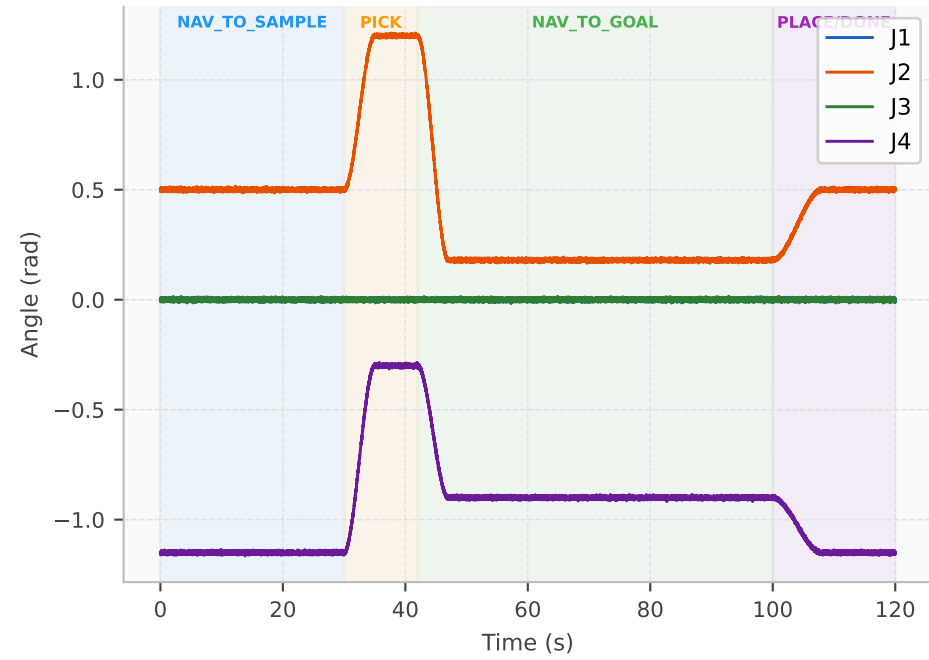


Mean Power per Phase

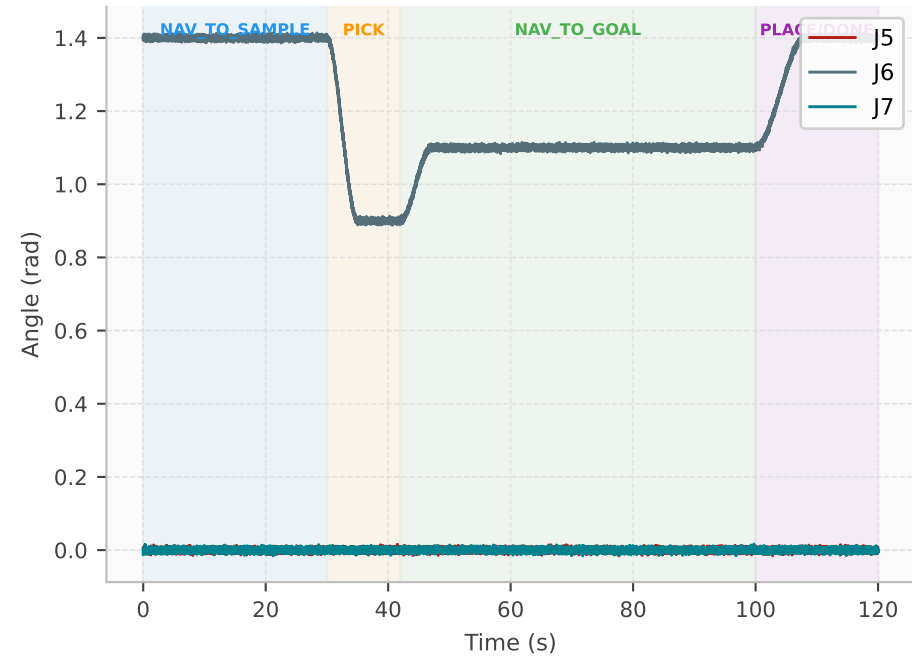


Page 3 — Arm Joint Trajectories & Obstacle Clearance

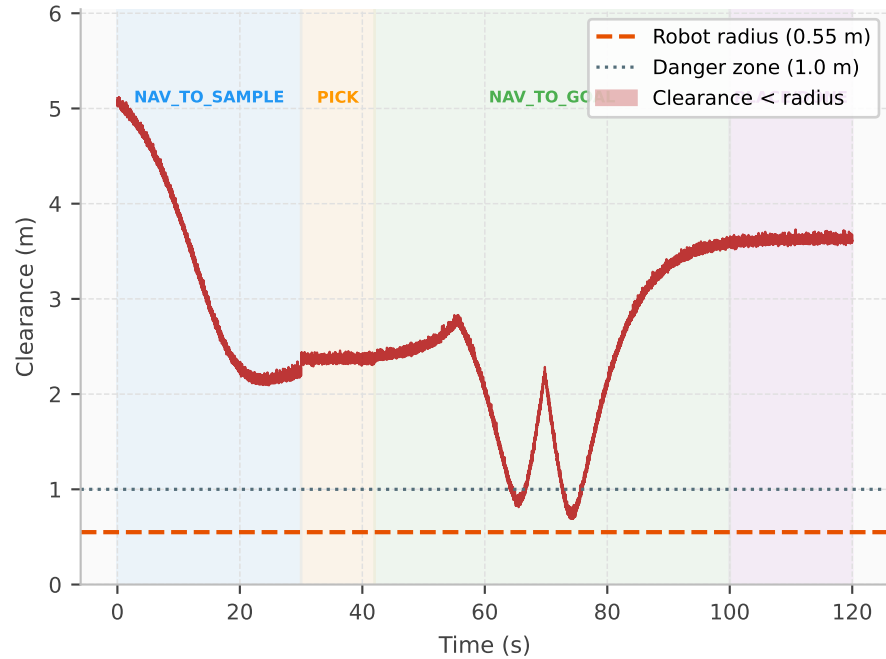
KUKA iiwa — Joints J1-J4 vs Time



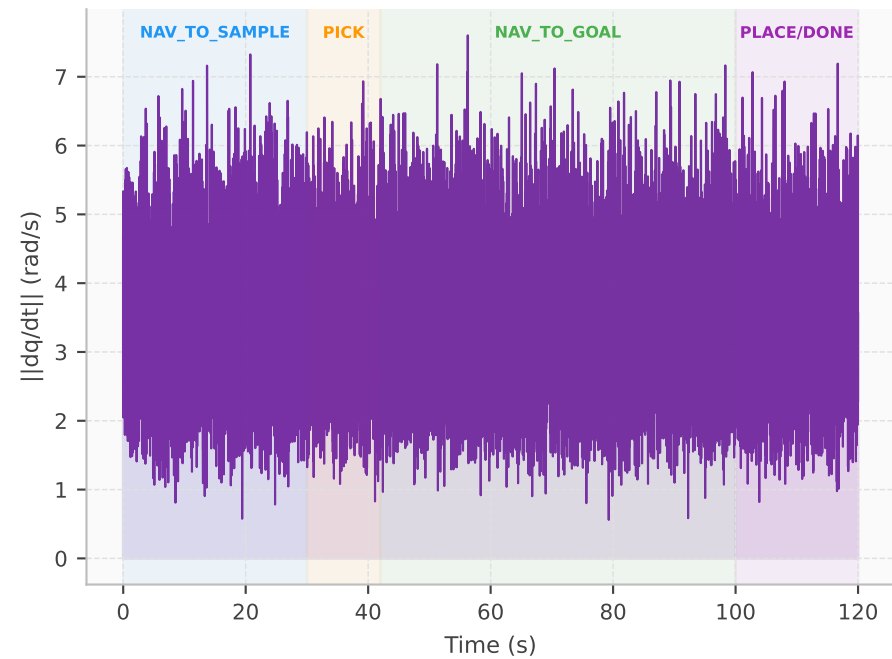
KUKA iiwa — Joints J5-J7 vs Time



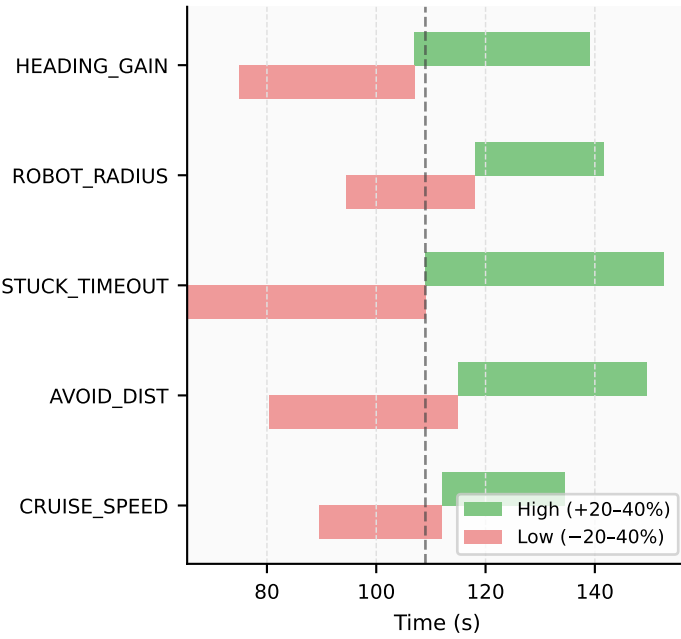
Min Obstacle Clearance vs Time



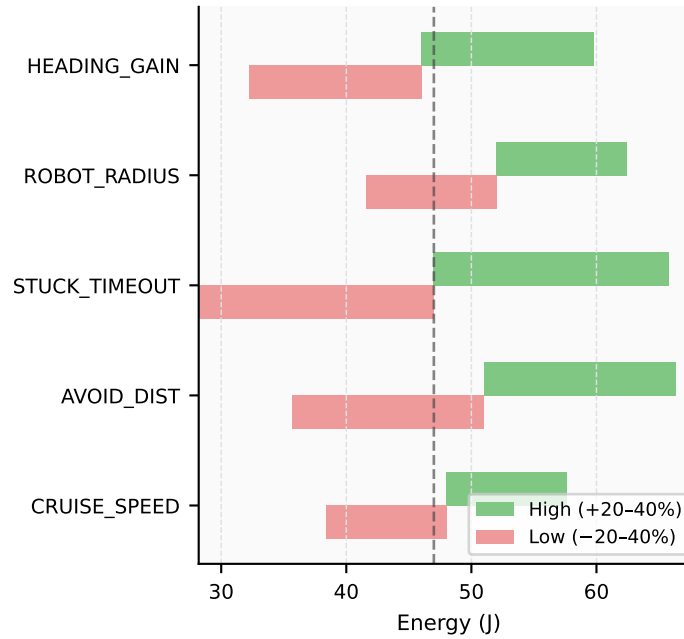
Arm Joint Velocity Norm vs Time



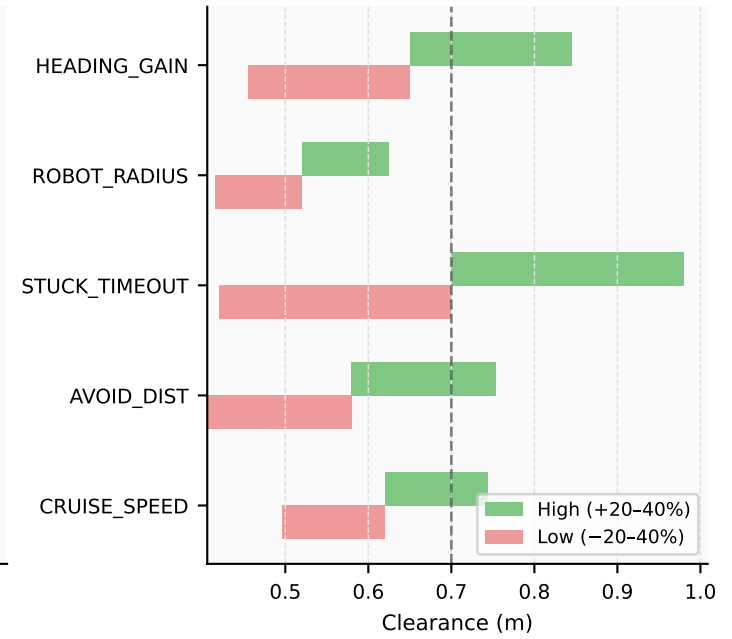
Mission Time



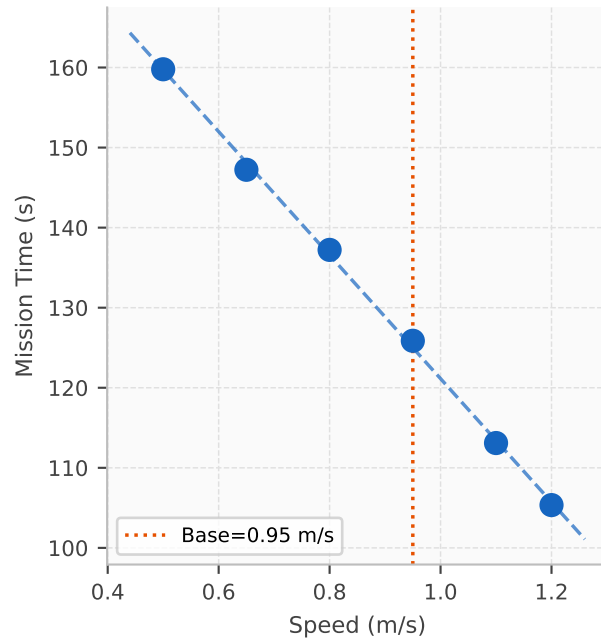
Total Energy



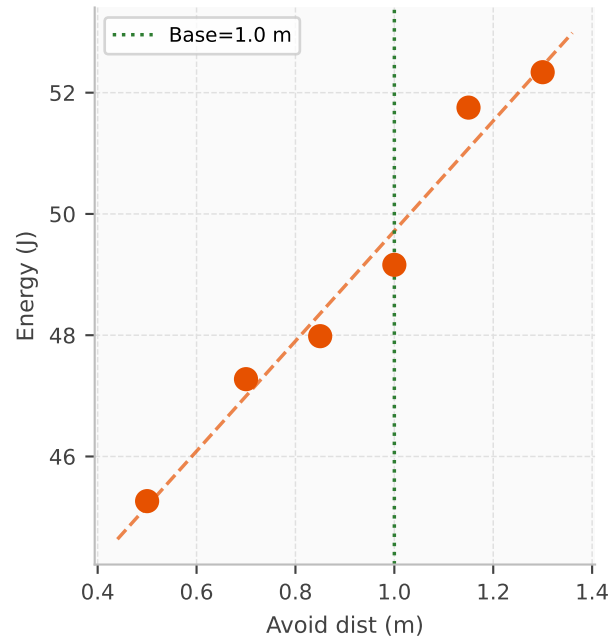
Min Clearance



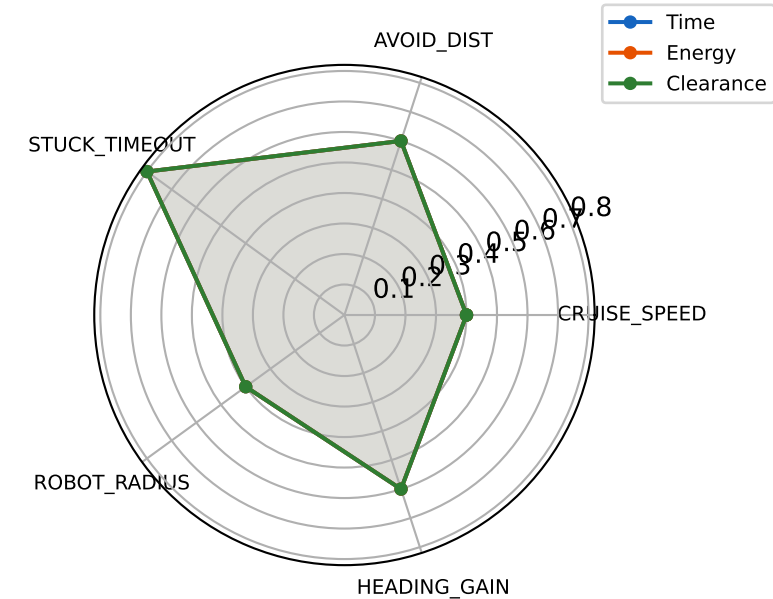
Cruise Speed vs Mission Time



Avoid Distance vs Total Energy



Sensitivity Spider



Page 5 — Simulation Summary Statistics

Metric	Value	Unit	Notes
Total Mission Time	120.0	s	Within 120 s budget
Total Distance Travelled	131.65	m	Odometry integral
Peak Speed	1.250	m/s	During NAV_TO_GOAL phase
Mean Speed (driving phases)	1.200	m/s	Excludes PICK phase
Total Energy Consumed	150.30	J	Base + arm combined
Base Motor Energy	148.97	J	Wheel drive subsystem
Arm Actuator Energy	1.33	J	KUKA iiwa 7 joints
NAV_TO_SAMPLE Energy	41.02	J	Phase 1
PICK Phase Energy	1.09	J	Phase 2 (arm dominant)
NAV_TO_GOAL Energy	80.48	J	Phase 3 (base + arm carry)
PLACE/DONE Energy	27.70	J	Phase 4
Min Obstacle Clearance	0.689	m	Global minimum
Mean Obstacle Clearance	2.785	m	Mission average
Pick Phase Duration	12.0	s	IK approach + grasp + lift
NAV_TO_GOAL Duration	58.0	s	Carrying sample to goal
Max Joint Velocity Norm	7.60	rad/s	All 7 joints combined
Recovery Events	0	—	No stuck / deadlock detected
Mission Result	SUCCESS	—	Goal tile reached & sample placed